

Unified Architecture Status Document

This document summarizes the agreed architectural direction after multiple external technical reviews and synthesis. It captures what has been validated, what is ready for implementation, and what specification work remains before coding begins.

1. System Vision

The platform is a low-cost, embeddable, open-source RAG chatbot engine allowing users to ingest content sources and deploy conversational interfaces powered by hybrid retrieval.

Primary goals:

- Cheap operation
- Easy embedding
- Multi-provider model flexibility
- Modular scalability path

2. Validated Architectural Decisions

The following design elements are considered stable and accepted:

- FastAPI-based stateless backend
- Managed PostgreSQL with pgvector
- S3-compatible object storage
- Dramatiq + Redis background processing
- Hybrid retrieval (vector + keyword)
- RRF fusion with reranking in pipeline
- iframe widget embedding approach
- External persistence and stateless compute philosophy
- Security baseline with short-lived tokens and origin checks
- Observability tooling (Sentry, Prometheus, tracing)

These choices passed multiple review cycles and are not expected to change.

3. Implementation-Ready Components

The system is architecturally ready for implementation in the following areas:

- High-level subsystem decomposition
- Data flow through ingestion and retrieval
- Deployment philosophy and scaling path
- Background job reliability model
- Retrieval pipeline structure
- Security direction and controls
- Observability integration strategy

At this stage, architecture redesign is not required.

4. Remaining Specification Work

Before production coding begins, the following implementation-level specifications must be completed to remove ambiguity:

1. Database schema definition
2. Retrieval pipeline parameter locking
3. Security policy detailing
4. Widget communication protocol
5. Cost modeling and projections
6. Observability SLO definition

These are documentation gaps rather than architectural flaws.

5. Current Readiness Position

The project status can be summarized as:

Idea Phase: Complete

Architecture Design: Complete

Technical Validation: Complete

Specification Drafting: In Progress

Implementation: Not started

Estimated remaining effort for specification completion:

3–5 focused days.

6. Required Next Documentation

The following documents must be authored next:

1■■ Database schema

2■■ Retrieval pipeline spec

3■■ Security spec

4■■ Widget protocol

5■■ Cost model

6■■ Observability SLOs

Completing these will enable confident development execution.

7. Conclusion

The architecture is stable and validated for a solo-developer deployment scenario. The remaining work is specification refinement rather than redesign. Once the above documentation tasks are complete, the project can move into implementation with low risk of architectural rework.